

CS-4110 HPC with GPUs

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C-205-E

Course Organization

Class communication

- Flex
- Google classroom

tbuf5mg

Course Intro

This course will teach the fundamentals needed to utilize the ever-increasing power of the accelerators, especially GPUs in an HPC system. The course will start with an architectural overview of modern HPC, especially GPU-based heterogeneous system, focusing on its computing power versus data movement needs. The course will cover a high level (pragma-based) GPU programming approach (Directive-based programming) as well utilization of existing libraries to quickly prototype an HPC application. Breadth of the course will cover lower-level approaches based on CUDA and OpenCL programming language for finer grained computationally intensive tasks. A particular attention will be given on performance tuning and techniques to overcome common data movement bottlenecks and patterns.

Course Learning Outcomes

1. **Describe** the terms related to HPC systems, GPU architecture and GPU programming models.
2. **Identify** application hotspots based on the application profile using realistic workload.
3. **Develop** data-parallel solutions using appropriate programming model.
4. **Analyze** performance of an application running on an HPC system to improve compute and memory performance.

Course Assessment

Assessment Item	Number	Weight (%)
Assignments	>3	10
Project	1	10
Sessional 1	1	15
Sessional 2	1	15
Quizzes	>4	10
Final Exam	1	40

Course Books

- Textbook: David B. Kirk, Wen-mei W. Hwu, “Programming Massively Parallel Processors A Hands-on Approach”, 3rd Edition (November 24, 2016), Elsevier.
1. Robert Robey, Yuliana Zamora , “Parallel and High Performance Computing”, 1st edition (July 2021), Manning Publications
 2. Aaftab Munshi, Benedict Gaster, Timothy G. Mattson, James Fung, Dan Ginsburg, OpenCL Programming Guide, 1st Edition, Publisher: Addison-Wesley Professional; 1st edition (July 23, 2011)
 3. Nvidia eBook, CUDA C++ Programming Guide, <https://docs.nvidia.com/cuda/cuda-c-programming-guide/index.html>
 4. Nvidia eBook, HPC for the Age of AI and Cloud Computing, <https://www.nvidia.com/en-us/lp/high-performance-computing/hpc-ai-cloud-computing-ebook/>

Software Tools

Languages: C, CUDA, Python

Tools: gcc, gprof, make, nvcc, nvprof

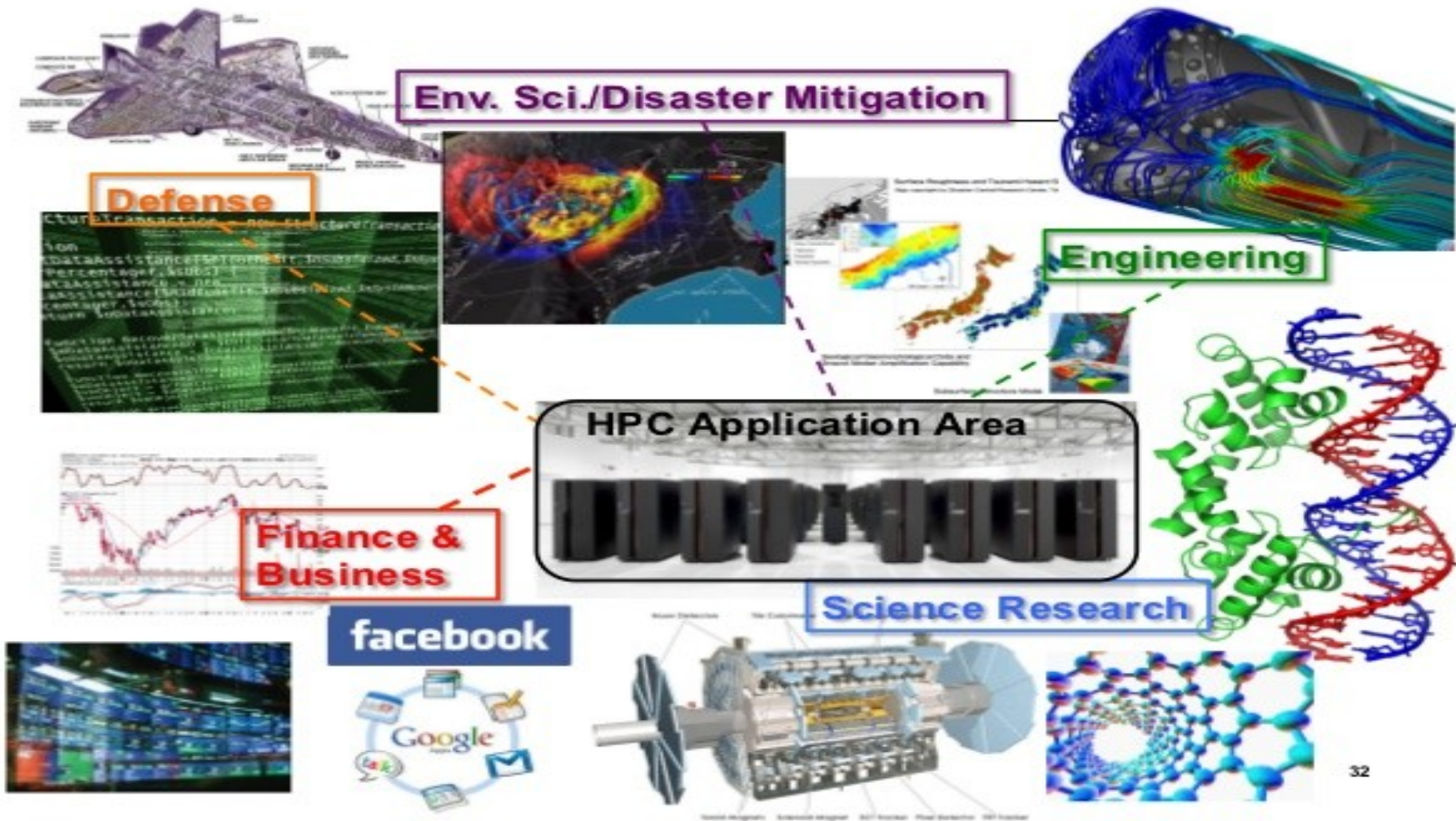
OS:Linux (Ubuntu)

Office hrs

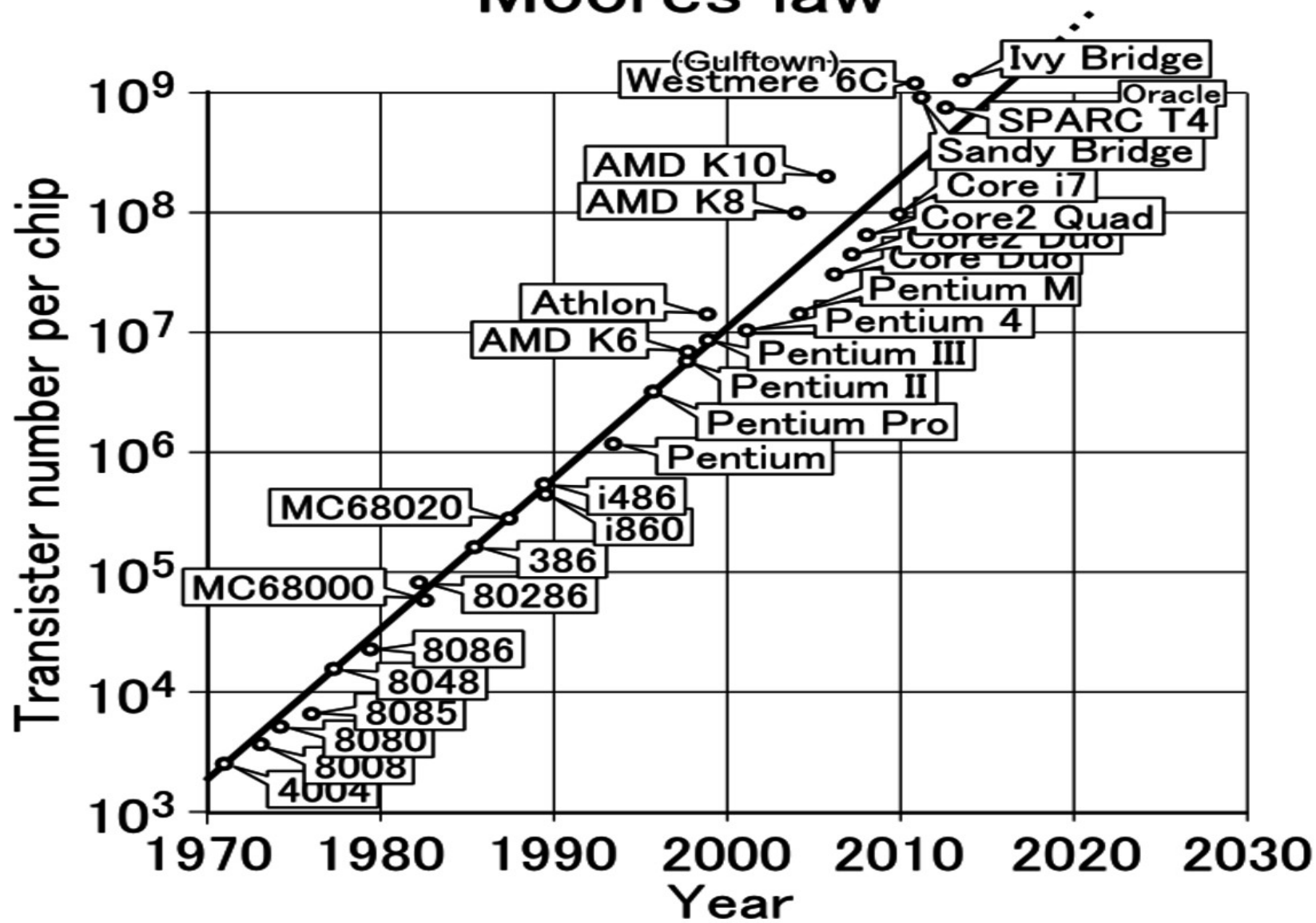
- Office: C-205-E
- Will be announced on GCR

Why HPC with GPUs?

High performance computing



Moore's law



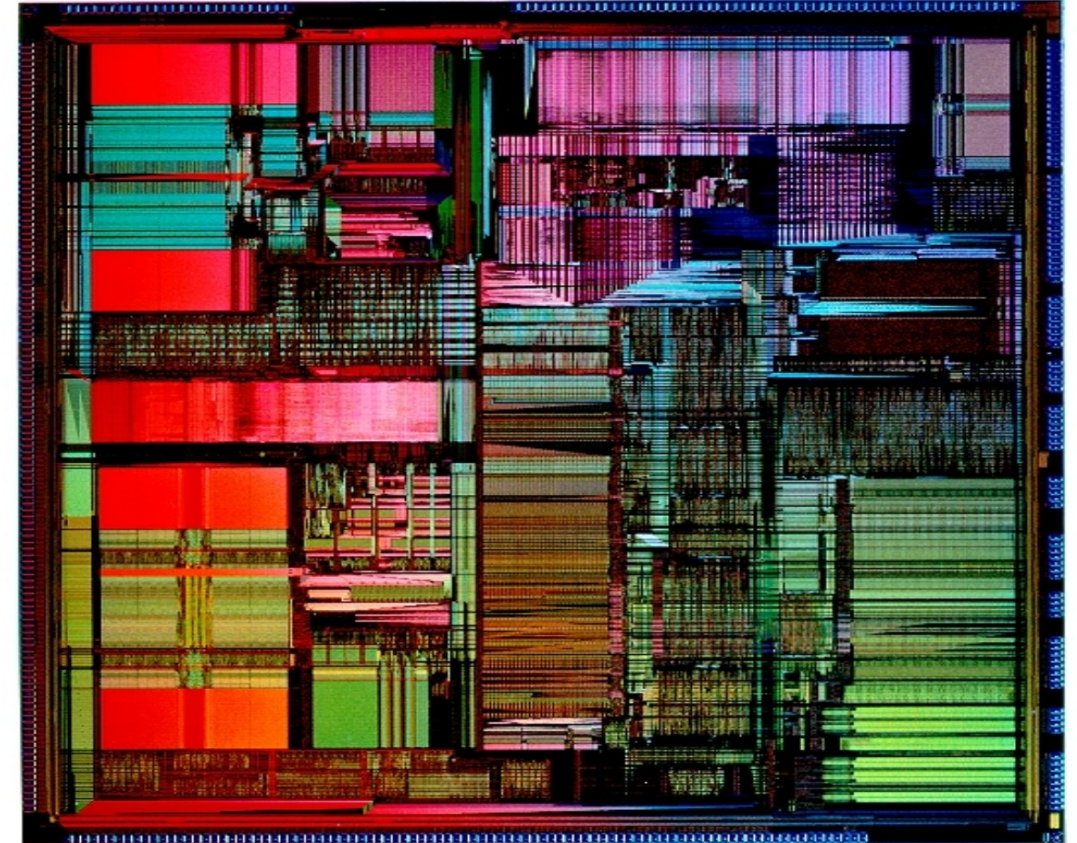
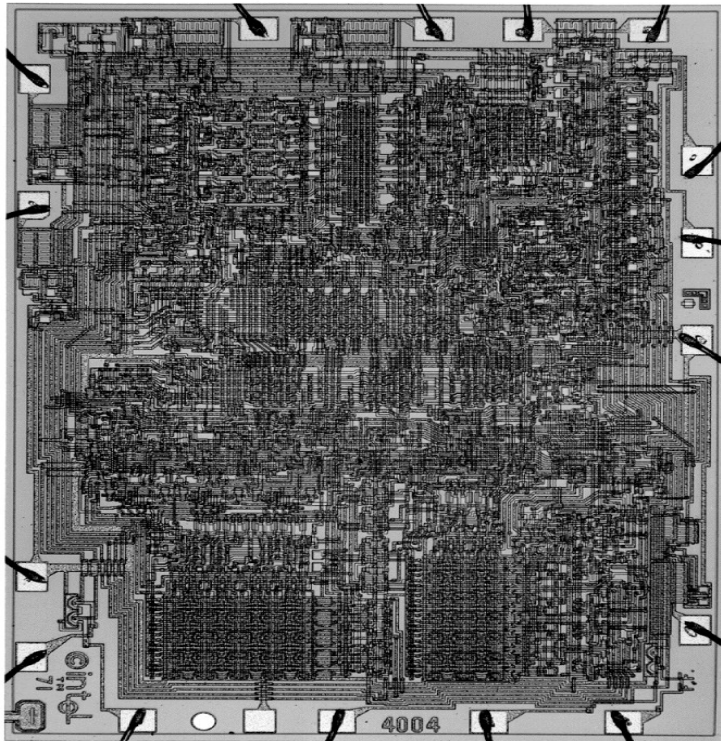
25 years of technology advance

INTEL 4004 (1971)

4-bit data

2300 transistors, 10 microns

0,06 MOPS, 108 kHz



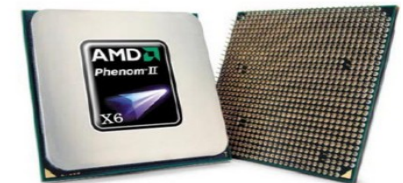
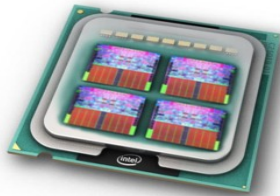
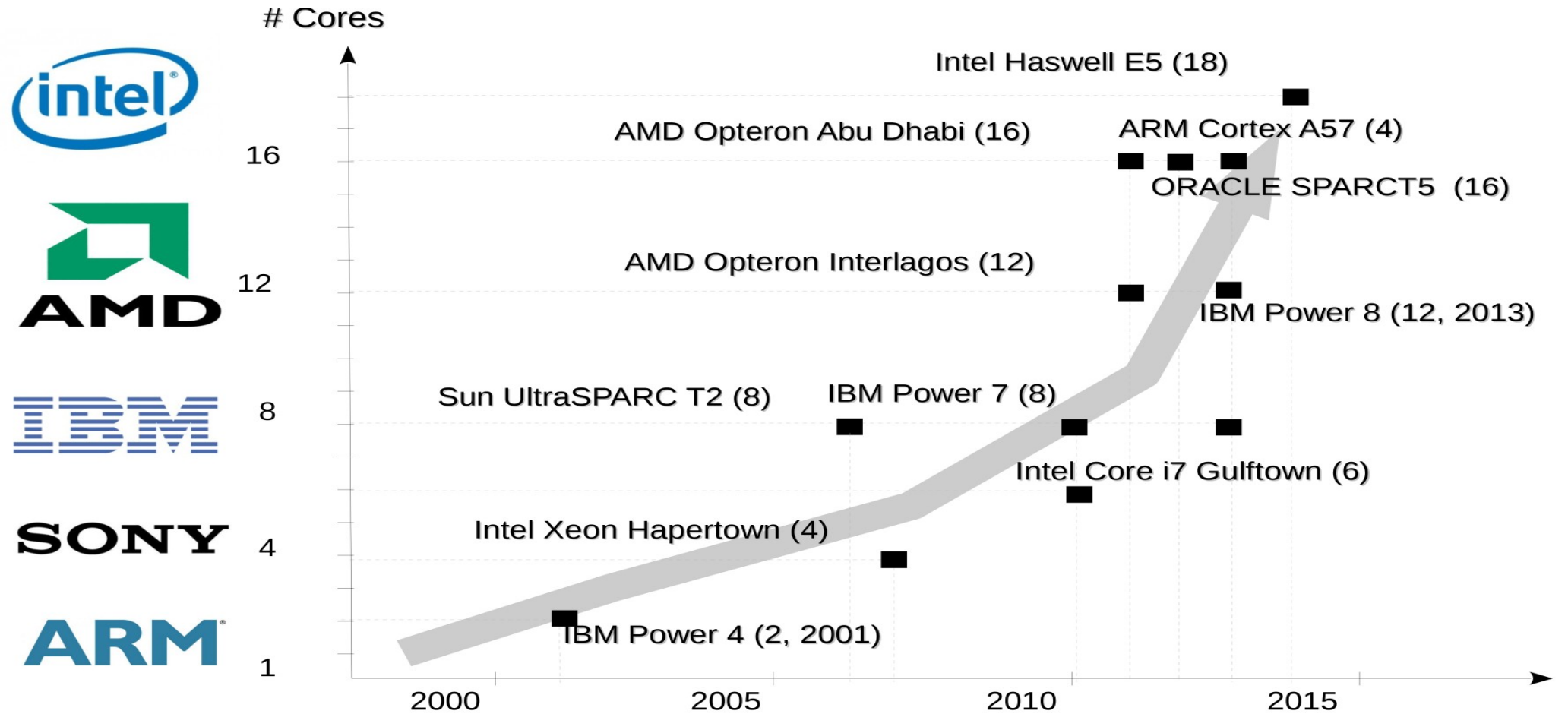
INTEL Pentium II (1996)

32-bit data

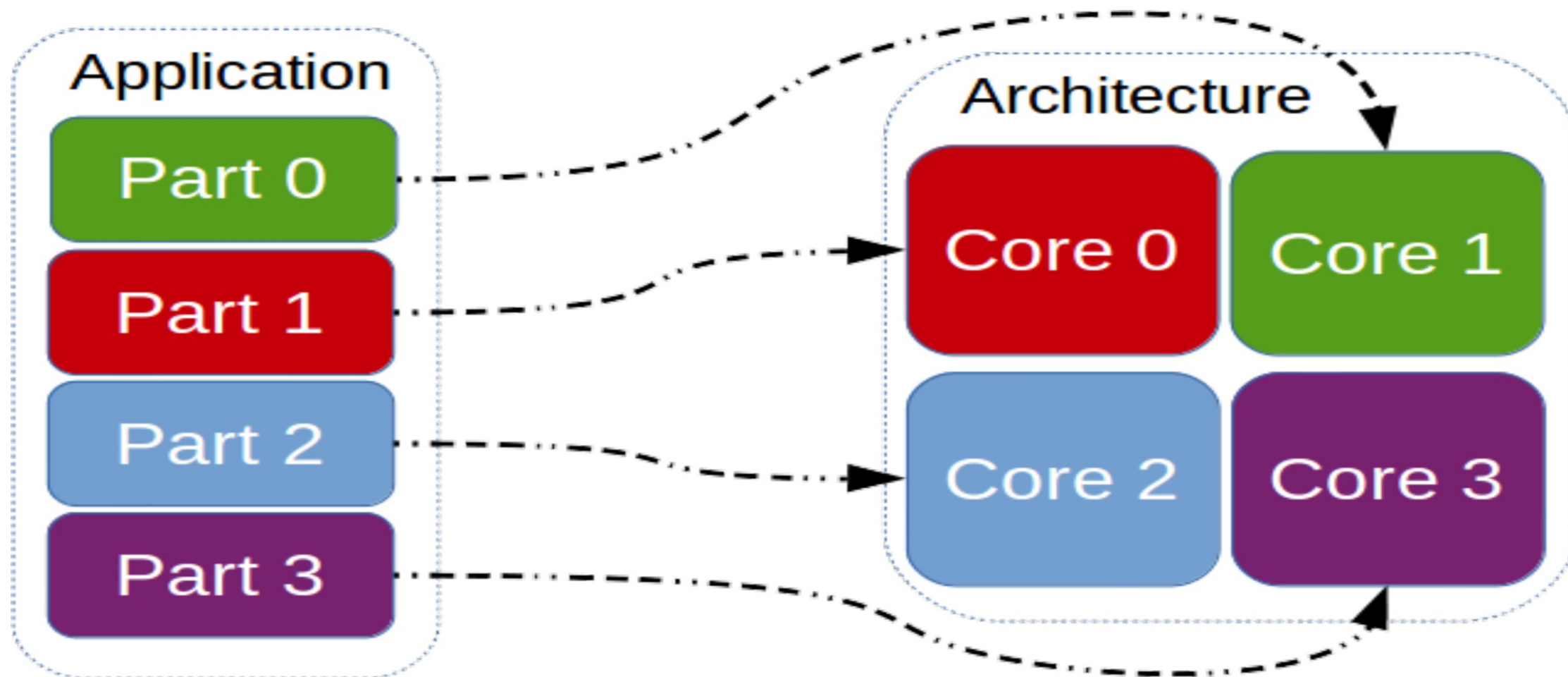
5.5M transistors, 0.35 μ , 2 cm²

200 MHz, 200 MOPS, 3.3V, 35W

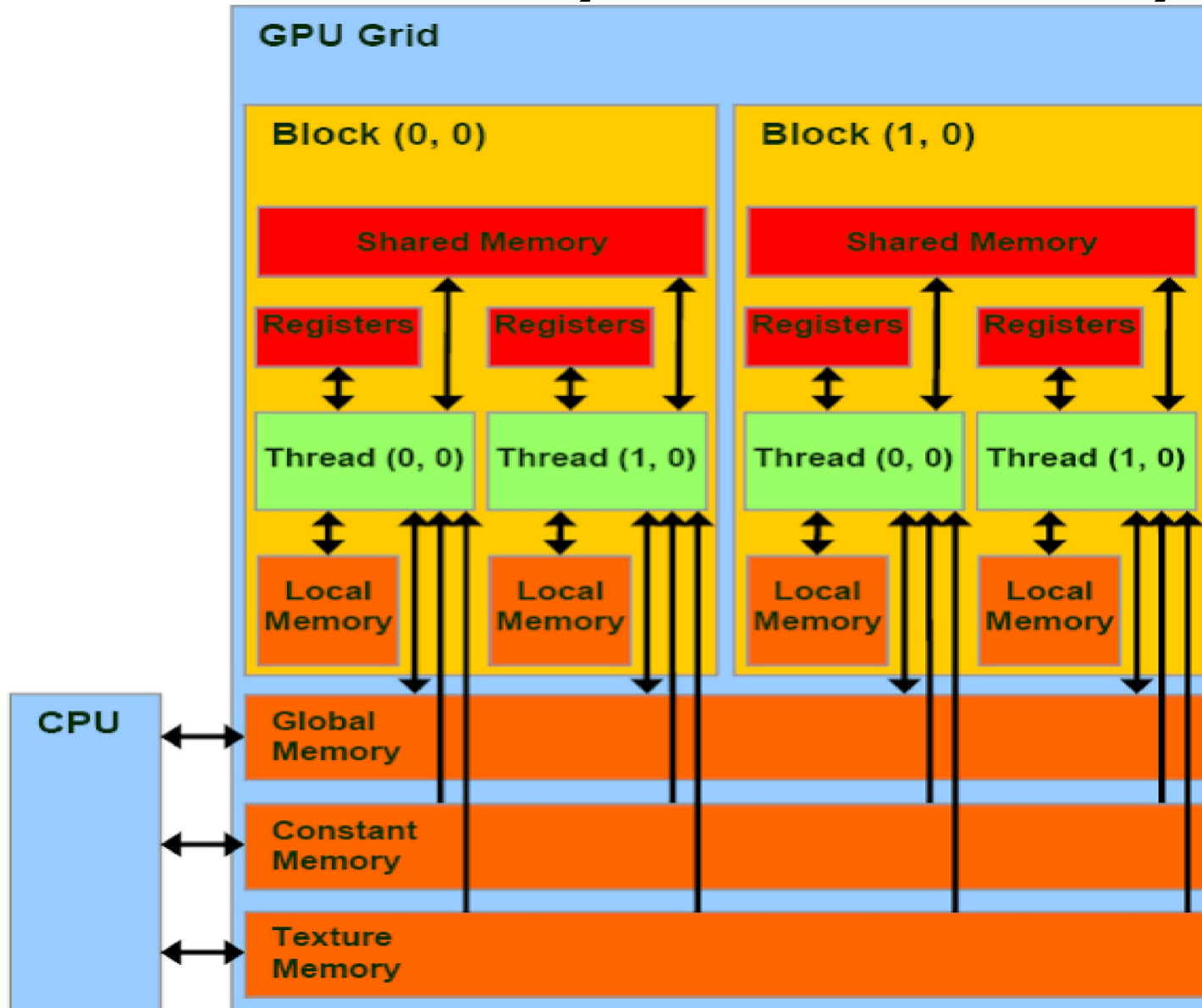
Multicore Trend



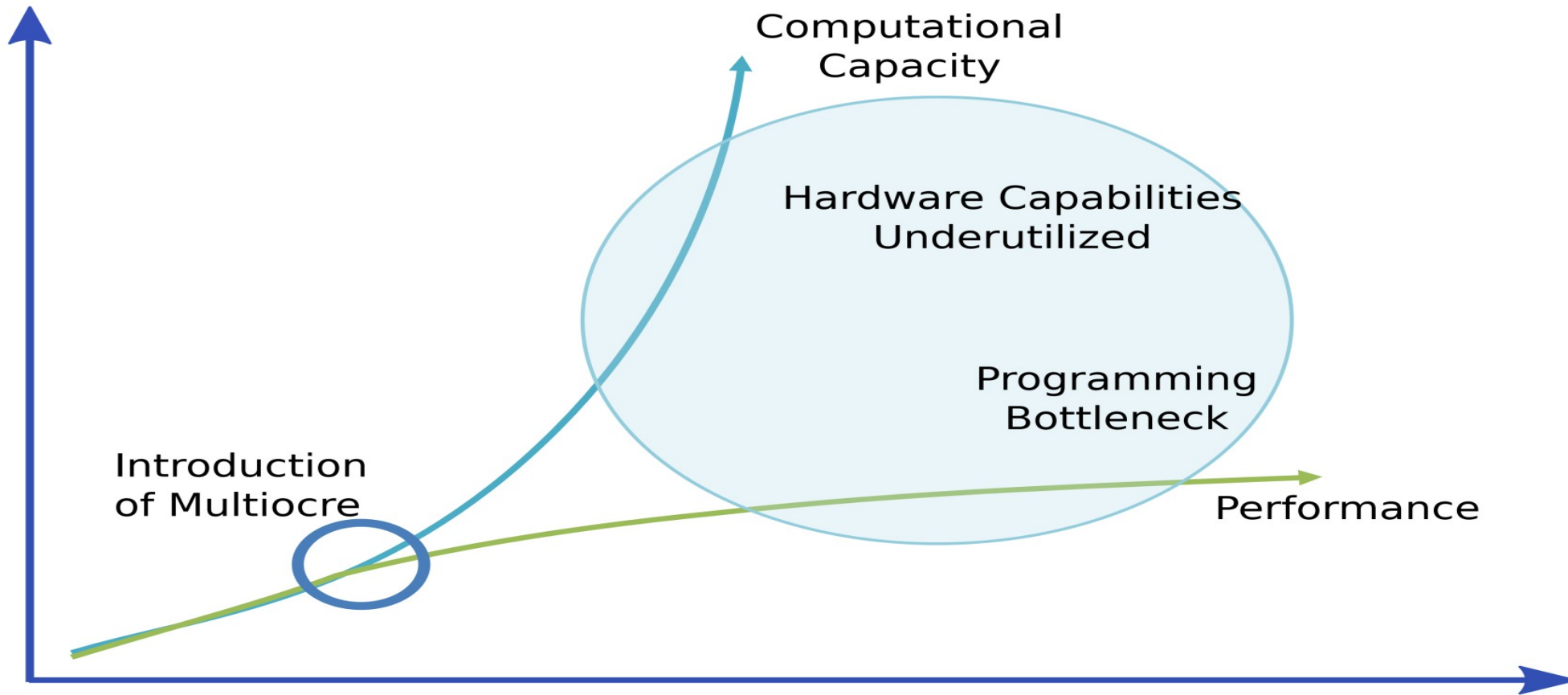
Application Partitioning



Memory Hierarchy



Productivity Gap



Computing Environment

- HPC System:
 - Details to be announced soon
- Terminals
 - Ubuntu
 - Your windows/Linux machines can also be used

Questions